Suppose you are conducting an experiment with two different types of bacteria, one in petri dish A (Dish A) at your laboratory and the other in petri dish B (Dish B). These two petri dishes are both in an isolated growth environment, and in the same laboratory. You also have four copper coins placed in any order either next to Dish A, or Dish B, or both (e.g., there may be one copper coin next to Dish A and three copper coins next to Dish B).

Over the course of a year-long experiment, you observe both dishes at 06:00 and 18:00 each day. At 06:00, if more growth had been experienced (during the last 12 hours) in **Dish A**, you pick up one copper coin that lies next to Dish A, and place it next to Dish B. If not, you do nothing. If there are no copper coins next to Dish A you also do nothing.

At 18:00, if more growth had been experienced (during the last 12 hours) in **Dish A**, you pick up one copper coin that lies next to Dish B, and place it next to Dish A. If not, you do nothing. If there are no copper coins next to Dish B you also do nothing.

(a) Model the outcome of your transfers of copper coins between the dishes as a Markov chain and produce the transition probability matrix of the chain.

(b) Let the fixed probability that Dish A will produce more growth during the last 12 hours be p. What is, in the long run, the probability that you will get to a point where you need to transfer a copper coin, but you cannot since there is no coin next to the dish that it needs to be transferred from.

(c) Let there now be N coins, how can the expression for the probability in question (b) be generalised to account for N?